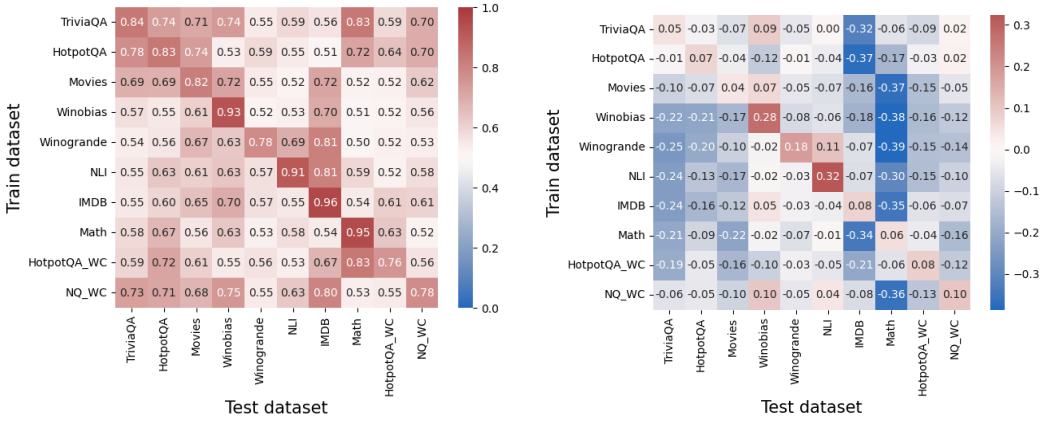


(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 9: Generalization between datasets, Llama-3-8b-instruct.

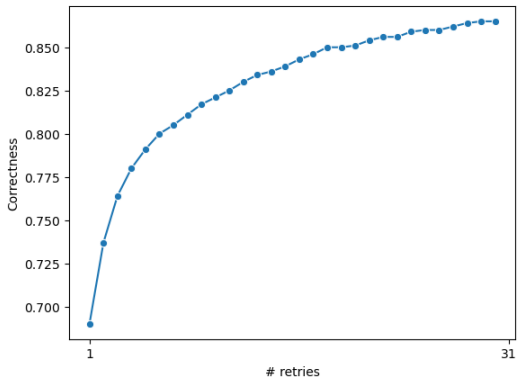


Figure 10: The percentage of answers for which at least one generated answer was correct. The first step is greedy decoding.

Table 8: AUC scores for error type classification (Winobias).

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	-	-	-	-
(B) Consistently correct	$0.83_{\pm 0.004}$	$0.88_{\pm 0.002}$	$0.84_{\pm 0.003}$	$0.89_{\pm 0.003}$
(C) Consistently incorrect	$0.83_{\pm 0.004}$	$0.88_{\pm 0.002}$	$0.79_{\pm 0.004}$	$0.90_{\pm 0.003}$
(D) Two competing	$0.68_{\pm 0.004}$	$0.58_{\pm 0.015}$	$0.74_{\pm 0.005}$	$0.88_{\pm 0.004}$
(E) Many answers	-	-	-	-

Table 9: AUC scores for error type classification (Math). Error types are predictable from the inner model representations, indicating the encoding of fine-grained information on errors.

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	-	-	-	-
(B) Consistently correct	$0.85_{\pm 0.017}$	$0.84_{\pm 0.007}$	$0.83_{\pm 0.020}$	$0.87_{\pm 0.006}$
(C) Consistently incorrect	$0.85_{\pm 0.026}$	$0.85_{\pm 0.003}$	$0.69_{\pm 0.032}$	$0.91_{\pm 0.007}$
(D) Two competing	-	$0.76_{\pm 0.020}$	$0.57_{\pm 0.001}$	$0.79_{\pm 0.006}$
(E) Many answers	$0.74_{\pm 0.010}$	$0.79_{\pm 0.015}$	$0.69_{\pm 0.041}$	$0.90_{\pm 0.008}$

D.1 Error Taxonomy Design Choices

The error taxonomy proposed in this paper is intentionally non-orthogonal, as some errors may simultaneously belong to multiple categories. For instance, an error might fall under both "consistently incorrect" (e.g., the same incorrect answer appears at least 15 times) and "many different answers" (e.g., the remaining answers show over 10 distinct variants).

Our taxonomy is designed to capture such nuanced cases, as restricting classification to a single category would hinder the generalizability of insights. Instead, we aim to learn general properties across different error types, providing LLM providers with actionable insights into questions exhibiting overlapping error patterns.

To support this non-orthogonal framework, our probes function as one-to-many classifiers, enabling precise error analysis and tailored solutions.

D.2 Results on Additional Datasets

Table 8 presents the results of error type classification on the Winobias dataset and Table 9 on the Math dataset.

D.3 Qualitative Examples

Tables 10 and 11 present qualitative examples of the error types in the TriviaQA and Math datasets.

E Detecting the Correct Answer Full Results

In Table 12 we present some qualitative samples from Mistral-7b-instruct, for the phenomenon we observe at error type **(C2)** Consistently incorrect but generates the correct answer at least one time. The samples in the table represent cases where the probe chose the correct answer. Table 13 compares different decoding mechanisms, including the choice via probe, on non-instruct models, and Table 14 compares on the instruct models. For all datasets and models, we observe similar conclusions to those in the main paper: significant improvement is observed for error types where the LLM shows no preference to the correct answer.

Table 10: Examples of error types in TriviaQA, Mistral-7B-Instruct. Correct answer is in **bold**.

Type of error	Question	Answers
Consistently correct	What clothing-part metaphorically classifies workers/jobs according to white or blue?	"**collar**": 30
Consistently incorrect	Which town in southeast Wales became a UNESCO World Heritage Site in 2000?	**"Blaenavon"**: 1, "Caerleon": 29
Many different answers	Published in 2013 who wrote the novel 'The Kill List'?	"**Frederick Forsyth**": 1, "Jerry Patterson": 1, "Edward Lee": 1, "Barry Lancet": 4 "Jeremy Holiday": 1, "Barry Lincoff": 1, "Jim Marrs": 1, "John Marrs": 1, "Anthony Lacy": 1, "Daniel Kraus": 1, "Ron Bass": 1, "David Martiniello": 2, "Eric Lustbader": 1, "Barbie Latza Nadeau": 1, "James Swallow": 1, "Mark Sullivan": 1, "Alex Binotto": 1, "David Baldacci": 1, "Bill Cosores": 1, "Frederic J. Brown": 1, "Ron Capps and Tate Foley": 1, "Barbie Wilde": 1, "NO ANSWER": 3
Two competing answers	What is the only letter of the alphabet which does not appear in any of the names of the 50 American states?	"**The letter q**": 15, "The letter X": 15

F Practical Guidance on Integrating Insights from this Paper into Model Development Workflows

The findings of this study reveal critical insights into the internal mechanisms of Large Language Models (LLMs) and their implications for truthfulness and error handling. To effectively incorporate these insights into model development, consider the following strategies:

Error Detection. Focus on representations of exact answer tokens to train the error detection probe. These tokens encode significant truthfulness signals and improve the reliability of error detection mechanisms. The trained probe should be integrated as part of the pipeline for specific task, e.g., math calculations. The probe provides a confidence score which can be used to warn the user for unreliable outputs, or to perform an intervention to fix the answer.

Error-Specific Interventions. The taxonomy of errors outlined in this study can be utilized to classify and analyze the types of errors that an LLMs may produce. Identifying these error types is useful for customizing strategies for error mitigation. The probes for detecting error types can be deployed as part of the LLM pipeline and create interventions based on their predictions. For example, Retrieval Augmented Generation (RAG) (Lewis et al., 2020) can help for "consistently incorrect"

Table 11: Examples of error types in Math, Mistral-7B-Instruct. Correct answer is in **bold**.

Type of error	Question	Answers
Consistently correct	If John travels 15 miles on a bike ride, and Jill travels 5 miles less, how many miles does Jim travel if he travels only 20% as far as Jill?	"**2**": 30
Consistently incorrect	Joy has 30 pencils, and Colleen has 50 pencils. If they bought the pencils at $4 each at the store, how much more money did Colleen pay than Joy for her pencils?	"**80$**": 1, "16$": 29 (correct)
Many different answers	If the first skyscraper was built 100 years ago, how many years in the future will it be 5 years before its 200th anniversary of being built?	"**95**": 14, "91": 1, "87": 1, "15": 2, "96": 1, "Six": 1, "202 ": 1, "2035": 1, "195": 1, "49": 1, "101": 1, "199": 1, "3 years before the 200th anniversary": 1, "203 years after it was built": 1, "196": 1, "2043": 1
Two competing answers	David did 27 more push-ups but 7 less crunches than Zachary in gym class today. If Zachary did 5 push-ups and 17 crunches.How many more crunches than push-ups did Zachary do?	"**12**":5, "1": 5 x

errors, as well as resampling and choosing the answer ranked highest by the error detection probe, or weight-update, if possible, as a more consistent solution. For "consistently correct" error types, an intervention on the LLM's internal representations can increase the confidence in generating a correct answer (Simhi et al., 2024).

Cross-Task Generalization. Universal generalization of probing classifiers across unrelated tasks should be approached with caution. The results in this work show that probes are mainly useful for task-specific error detection.